

AAKASH PAI

BSc (Hons) Computer Science Student | Kuala Lumpur, Malaysia

INTERNSHIP PERIOD: JAN 2027 – APRIL 2027

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LOCATION: Mont Kiara, Kuala Lumpur

OWN TRANSPORTATION: Yes



Career Objective

Computer Science undergraduate at Sunway University (Lancaster University Dual Award) specializing in backend systems, distributed architecture, and concurrent programming in Java and Spring Boot. Seeking a Software Engineering Internship for the January – April 2027 cohort to build high-throughput backend infrastructure.

Education

Sunway University & Lancaster University (Dual Award), Malaysia

Sep 2025 – Sep 2027
(Expected)

BSc (Hons) Computer Science

- **CGPA:** 3.54 / 4.00
- **Dean's List Award** – February 2026 Semester, Faculty of Engineering and Technology
- **ACE Scholarship** – RM 11,000/year merit scholarship based on academic performance

Sunway College, Sunway City, Selangor, Malaysia

Aug 2023 – Aug 2025

Diploma in Information Technology

- **CGPA:** 3.83 / 4.00
- **Relevant Coursework:** Web Development, Database Fundamentals, Mobile Application Development, Object-Oriented Modeling, Java Programming, Systems Analysis & Design, Computer System Architecture

Work Experience

Frontend Developer Intern, InQuantum AI, Kuala Lumpur, Malaysia

Apr 2025 – Aug 2025

- Built core UI modules and responsive dashboard views for xPulse, an internal AI analytics platform, using Next.js, TypeScript, and Tailwind CSS.
- Integrated backend REST APIs with end-to-end state management, request caching, and error boundaries to ensure interface reliability.
- Automated internal cross-service data workflows using n8n and developed production UI micro-interactions using Framer Motion.

Technical Projects

Distributed API Gateway & Throttling Core |

May 2026

github.com/kashals/distributed-rate-limiter-gateway

- Built a high-throughput API gateway in pure Go that enforces per-user rate limits globally across horizontally scaled replicas with throttling state managed in Redis.
- Implemented pluggable Token Bucket (lazy refill via HSET) and Sliding Window Log (ZSET) algorithms via atomic Lua scripts in Redis to eliminate race conditions.
- Implemented JWT middleware supporting HS256 and RS256; verified identity is injected as X-User-ID downstream while stripping raw Authorization headers before proxying.

- Benchmarked gateway cluster using k6, sustaining over 10,000 req/sec across local replicas with sub-5ms p99 latency without Redis connection pool exhaustion.
- **Stack:** Go, Redis, Lua

Sugar And Icing | sugarandicing.vercel.app | github.com/kashals/sugar-and-icing Jan 2025 – July 2025

- Production e-commerce platform for a local bakery featuring server-authoritative Stripe checkout, Cloudflare Turnstile bot verification, and Upstash Redis rate limiting.
- Implemented atomic inventory deduction via Supabase PostgreSQL Remote Procedure Calls (RPC) to prevent overselling race conditions under concurrent checkouts.
- Built a role-gated administrative analytics dashboard tracking live store sales and order fulfillment state.
- **Stack:** Next.js 15, TypeScript, Supabase, Stripe, Upstash Redis, Tailwind CSS

Order Settlement Engine | github.com/kashals/order-settlement-engine Oct 2026

- Architected an asynchronous event-driven settlement engine across two Spring Boot 3 microservices communicating via Apache Kafka with isolated PostgreSQL databases.
- Implemented the Transactional Outbox pattern using database row-level locking to resolve dual-write inconsistencies and prevent duplicate message dispatch across service replicas.
- Engineered atomic consumer deduplication using relational database constraints, safely discarding re-delivered Kafka events without triggering Hibernate session rollback pollution.
- Built resilient consumer error handling with exponential backoff retries and dead-letter queue routing to isolate malformed payloads without blocking partition processing.
- Benchmarked ingestion throughput using hey, sustaining 254 req/sec at 77ms average latency (p99 200ms) with zero message drops and verified consistency across terminal saga states.
- **Stack:** Java 21, Spring Boot 3, Apache Kafka, PostgreSQL, Docker, Flyway

Core Competencies & Technical Skills

Languages: Go, TypeScript, JavaScript, Python, Java, SQL, Lua

Backend & Distributed Systems: Node.js, Express, REST APIs, Redis, Concurrency, JWT Authentication, Lua Scripting, Microservices

Databases & Cloud Infrastructure: PostgreSQL, Supabase, Docker, Google Cloud Run, Git, Upstash

Frontend: Next.js, React, Tailwind CSS, HTML5, CSS3, Framer Motion

Extracurricular Activities & Achievements

- Participated in **MyHack 2026** (GDG Hackathon) at Sunway University, building an AI-powered venture decision-support system within a strict 24-hour timeframe.
- Active member of the **Sunway Cybersecurity Club**, Sunway University, engaging in technical workshops on web vulnerabilities and system security.